

Handwritten Recognition Game

Course Name: Software Development

Programming Language: Java

Database: Not Required

10-Week Work Plan

Week 1

Just To Do:

- Understand the project requirements
- Decide the game objectives and features
- Install Java JDK and IDE
- Create the project folder structure
- Create a GitHub repository

To Upload:

- Project setup files
- Initial project structure
- Weekly progress report

Week 2

Just To Do:

- Learn Java Swing basics
- Learn event handling concepts
- Create the main application window
- Add buttons, labels, and panels
- Design the Home Page

To Upload:

- Home Page source code
- Screenshots of the Home Page
- Weekly progress report

Week 3

Just To Do:

- Create the Instruction Page
- Create the Game Page layout
- Create the Result Page
- Connect all pages using navigation buttons
- Test page transitions

To Upload:

- GUI source code
- Screenshots of all pages
- Weekly progress report

Week 4

Just To Do:

- Develop the drawing canvas
- Implement mouse input handling
- Allow users to draw letters and numbers
- Add Clear button functionality
- Test drawing operations

To Upload:

- Drawing canvas source code
- Canvas screenshots
- Updated project files

Week 5

Just To Do:

- Add Submit button functionality
- Capture user handwriting input
- Process drawing data
- Store temporary input for recognition
- Test input collection system

To Upload:

- Input handling module
- Test screenshots
- Weekly progress report

Week 6**Just To Do:**

- Implement character recognition logic
- Compare user input with predefined patterns
- Display recognition results
- Improve recognition accuracy
- Test recognition functionality

To Upload:

- Recognition module source code
- Recognition test results
- Weekly progress report

Week 7**Just To Do:**

- Develop the scoring system
- Add timer functionality
- Generate random characters for users
- Display success and error messages
- Connect scoring with recognition results

To Upload:

- Game logic source code
- Feature screenshots
- Updated README file

Week 8**Just To Do:**

- Improve the game interface
- Enhance recognition accuracy
- Improve user experience
- Optimize application performance
- Refine game flow and feedback system

To Upload:

- Updated source code
- Updated screenshots
- Weekly progress report

Week 9

Just To Do:

- Perform complete system testing
- Identify bugs and errors
- Fix functionality issues
- Improve application stability
- Verify all game features

To Upload:

- Testing report
- Bug-fix documentation
- Updated project files

Week 10

Just To Do:

- Complete project documentation
- Organize source code
- Update GitHub repository
- Prepare final presentation slides
- Review the complete project before submission

To Upload:

- Complete Java source code

- Final README file
- Project report
- Presentation slides
- Final screenshots
- Complete GitHub repository

Step 1: What I Will Do

For this project, I will develop a Handwritten Recognition Game using Java. First, I will analyze the project requirements and plan the overall structure of the application. Then, I will create a graphical user interface where users can interact with the game and write characters on a drawing canvas. After that, I will implement a handwritten recognition system that can identify the user's input by comparing it with predefined character patterns. I will also develop game features such as scoring, timer functionality, random character generation, and feedback messages. Finally, I will test the application, fix any issues, and improve its performance before the final submission.

Step 2: What I Need to Learn

To complete this project successfully, I need to strengthen my knowledge of Java programming and object-oriented programming concepts. I will learn how to create graphical user interfaces using Java Swing and how to handle mouse and button events. Since the project involves handwritten character recognition, I also need to understand basic image processing concepts, pattern matching techniques, and character recognition methods. In addition, I will learn debugging, software testing, and GitHub version control to manage the project effectively.

Step 3: What I Will Submit on GitHub

I will upload the complete Java source code of the project to GitHub. The repository will contain all project files, documentation, screenshots, testing reports, and progress updates. A README file will explain the project objectives, features, tools used, installation process, and instructions for running the application. Regular commits will be maintained throughout the development process to show project progress. The final repository will serve as the complete record of the finished project.